

TweetCrisis Analyser - Disaster and Humanitarian Category Classification from Tweets using Transformer Models

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Abstract. Automated analysis of social media during disasters helps responders quickly extract valuable information. We study a dual classification task on the HumAID dataset: (1) identifying the *disaster type* and (2) predicting the *humanitarian category* in tweets. We fine-tune a multi-head Transformer based on BERTweet and evaluate it on HumAID. Our model achieves **99%** accuracy for disaster type and **78.6%** for humanitarian-category classification. We present dataset statistics, model architecture, training details, error analysis, and directions for future work.

Keywords: HumAID · BERTweet · Transformer · Disaster Classification · Humanitarian Labels · Multi-task Learning

1 Introduction

Disasters, natural or human-made, generate large volumes of social media posts. Platforms like Twitter often provide timely, on-ground information such as rescue needs, damage reports, and offers of assistance. The volume and speed of posts make manual monitoring impractical for emergency response teams. Automated classification systems can help extract meaningful insights quickly. Emergency organizations need tools to identify two key aspects from a tweet:

- **Disaster type being discussed**
- **Humanitarian need or action expressed**

We address both tasks using a **dual classification** framework that predicts disaster type and humanitarian category from a single tweet.

Problem Statement: Given a tweet, automatically predict:

1. The **disaster type** (e.g., Flood, Earthquake, Wildfire)
2. The **humanitarian category** (e.g., Urgent Need, Donation, Search & Rescue, Volunteer, Other)

2 Related Work

Crisis informatics leverages social media to support disaster response by extracting situational awareness. Resources like CrisisNLP have facilitated research on identifying disaster-related content.

The HumAID dataset provides human-annotated tweets labeled for *disaster type* and *humanitarian category*, serving as a standard benchmark for dual-task classification [1].

Transformer models, with their self-attention mechanism and strong contextual representations, are widely used in NLP [3]. BERTweet, pre-trained on a large tweet corpus, adapts well to noisy social media text [2].

Building on these, we fine-tune BERTweet with a multi-head classification setup to jointly predict disaster type and humanitarian category.

3 Dataset Description

3.1 Dataset Source

We use the HumAID dataset [1], released by Alam et al. (2021) via the [CrisisNLP Project](#). HumAID is one of the largest human-annotated Twitter datasets for crisis informatics.

3.2 Annotation Scheme

HumAID provides annotations for:

- **Disaster Type:** e.g., Flood, Wildfire, Earthquake, Hurricane, Cyclone etc.

- **Humanitarian Label:** e.g., Urgent Need, Volunteer, Donation, Search & Rescue, Infrastructure Damage, Other Relevant Information etc.

3.3 Dataset Statistics

Combining the training and development splits gives 33,006 labeled tweets for training, covering multiple real-world disaster events.

Table 1 provides an overview of the dataset, including sources, modalities, and coverage.

Table 1: Total Dataset Overview

Aspect	Description
Source	CrisisNLP
Modality	Twitter Text Data
Total Tweets (Train + Dev)	33,006
Total Tweets (Test)	8,161
Number of Disaster Types	5
Number of Humanitarian Categories	10
Language	English
Number of Distinct Locations	12
Period Covered	2016–2019

Table 2 shows the distribution of disaster types in the train and test sets, highlighting class imbalance.

Table 2: Disaster Type Distribution in Train+Dev and Test Sets

Disaster Type	Train (N=33,006)		Test (N=8,161)	
	Count	Percentage	Count	Percentage
Hurricane	22005	66.7%	5438	66.7%
Earthquake	5590	16.9%	1384	17.0%
Cyclone	3154	9.6%	779	9.6%
Wildfire	1214	3.7%	301	3.7%
Flood	1043	3.1%	259	3.2%

Table 3 presents the distribution of humanitarian labels across train and test splits.

Table 3: Humanitarian Label Distribution in Train+Dev and Test Sets

Humanitarian Label	Train (N=33,006)		Test (N=8,161)	
	Count	Percentage	Count	Percentage
Rescue Volunteering or Donation Effort	9003	27.3%	2227	27.3%
Other Relevant Information	5878	17.8%	1455	17.8%
Infrastructure and Utility Damage	4647	14.1%	1147	14.1%
Sympathy and Support	3944	12.0%	976	12.0%
Injured or Dead People	3396	10.3%	840	10.3%
Caution and Advice	1748	5.3%	433	5.3%
Not Humanitarian	1531	4.6%	376	4.6%
Displaced People and Evacuations	1409	4.3%	347	4.3%
Requests or Urgent Needs	1307	4.0%	324	4.0%
Missing or Found People	143	0.4%	36	0.4%

Tables 4 and 5 detail label distributions across locations and year–disaster types, useful for analyzing event-specific trends.

Table 4: Humanitarian and Disaster Labels across Different Locations

Humanitarian Label	Split	Ecuador	Greece	Harvey	Idai	Irma	Italy	Kaikoura	Maria	Maryland	Matthew	Puebla Mexico	Sri Lanka
Rescue Volunteering or Donation Effort	Train	316	285	2264	1499	1275	250	166	1586	58	261	788	255
	Test	78	71	559	370	315	62	41	391	15	65	196	64
Other Relevant Information	Train	65	127	1417	326	1891	41	250	1257	110	262	87	45
	Test	16	32	350	81	467	11	61	311	27	66	22	11
Infrastructure and Utility Damage	Train	56	31	976	284	1509	43	250	1144	63	143	134	14
	Test	14	7	241	70	372	11	62	283	16	35	33	3
Sympathy and Support	Train	256	258	509	387	455	464	346	539	88	316	294	32
	Test	63	64	126	95	112	115	86	133	22	79	73	8
Injured or Dead People	Train	445	396	559	347	717	140	84	242	45	180	204	37
	Test	110	99	139	86	177	34	21	60	11	44	50	9
Displaced People and Evacuations	Train	0	0	552	46	605	0	70	105	0	31	0	0
	Test	0	0	136	11	150	0	17	26	0	7	0	0
Caution and Advice	Train	24	21	434	71	491	0	395	176	56	29	28	23
	Test	6	5	107	18	122	0	98	44	14	7	7	5
Requests or Urgent Needs	Train	73	20	267	115	101	24	19	570	0	42	49	27
	Test	18	5	66	28	25	6	5	141	0	11	12	7
Not Humanitarian	Train	18	60	329	64	493	0	180	217	62	61	31	16
	Test	5	14	81	16	122	0	44	53	15	15	7	4
Missing or Found People	Train	0	16	0	15	0	0	0	0	112	0	0	0
	Test	0	4	0	4	0	0	0	0	28	0	0	0
Total	Train	1253	1214	7307	3154	7537	962	1760	5836	594	1325	1615	449
	Test	310	301	1805	779	1862	239	435	1442	148	329	400	111

Table 5: Humanitarian Labels across Different Year–Disaster Types

Humanitarian Label	Split	2016 Earthquake	2016 Hurricane	2017 Earthquake	2017 Flood	2017 Hurricane	2018 Flood	2018 Wildfire	2019 Cyclone
Rescue Volunteering or Donation Effort	Train	732	261	788	255	5125	58	285	1499
	Test	181	65	196	64	1265	15	71	370
Other Relevant Information	Train	356	262	87	45	4565	110	127	326
	Test	88	66	22	11	1128	27	32	81
Infrastructure and Utility Damage	Train	349	143	134	14	3629	63	31	284
	Test	87	35	33	3	896	16	7	70
Sympathy and Support	Train	1066	316	294	32	1503	88	258	387
	Test	264	79	73	8	371	22	64	95
Injured or Dead People	Train	669	180	204	37	1518	45	396	347
	Test	165	44	50	9	376	11	99	86
Displaced People and Evacuations	Train	70	31	0	0	1262	0	0	46
	Test	17	7	0	0	312	0	0	11
Caution and Advice	Train	419	29	28	23	1101	56	21	71
	Test	104	7	7	5	273	14	5	18
Requests or Urgent Needs	Train	116	42	49	27	938	20	20	115
	Test	29	11	12	7	232	5	5	28
Not Humanitarian	Train	198	61	31	16	1039	62	60	64
	Test	49	15	7	4	256	15	14	16
Missing or Found People	Train	0	0	0	0	0	112	16	15
	Test	0	0	0	0	0	28	4	4
Total	Train	3975	1325	1615	449	20680	594	1214	3154
	Test	984	329	400	111	5109	148	301	779

3.4 Data Preprocessing

The HumAID dataset required several preprocessing steps before model training, as summarized below:

- Cleaned raw tweets and removed duplicates.
- Normalized text while keeping hashtags, mentions, and emojis.
- Encoded labels using scikit-learn `LabelEncoder`.
- Tokenized with BERTweet tokenizer in case of BERTweet Model and Qwen2 tokenizer in case of Qwen2-1.5B Model (max length 128, padding applied).
- Used official train+dev splits for fine-tuning; test split for evaluation.

4 Methodology - BERTweet Transformer Model

4.1 Overview

We use a **multi-task learning** framework based on **BERTweet** to jointly classify tweets into disaster types and humanitarian categories. A shared BERTweet encoder produces contextual embeddings, which are fed into two **task-specific classification heads**. The model is trained using a **combined cross-entropy loss**.

4.2 System Pipeline

Figure 1 shows the workflow. Tweets are cleaned and tokenized, passed through the shared BERTweet encoder, and processed by two heads: *Disaster Head* and *Humanitarian Head*, trained jointly with a combined loss.

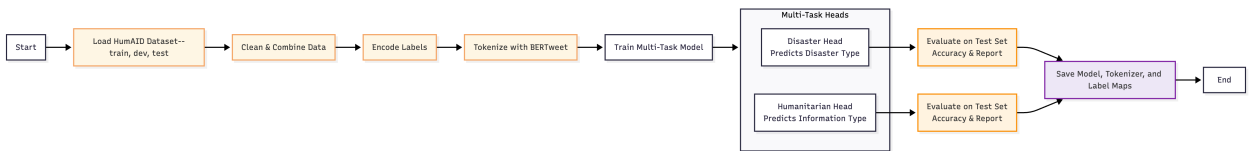


Fig. 1: Multi-task tweet classification pipeline using BERTweet.

4.3 Model Architecture

We use `vinai/bertweet-base` with a dual-head design. The 768-dim encoder embeddings feed into two linear layers (one per task) with dropout 0.2. The total loss is the sum of the cross-entropy losses for both tasks:

$$\mathcal{L} = \mathcal{L}_{\text{disaster}} + \mathcal{L}_{\text{human}}$$

to encourage shared representation learning while preserving task-specific boundaries.

5 Methodology - Qwen2-1.5B Fine-Tuning with LoRA

5.1 Overview

We fine-tuned the `Qwen2-1.5B` decoder-only model using *Low-Rank Adaptation (LoRA)*, a parameter-efficient method that updates less than 0.1% of the parameters. This allowed the model to learn crisis-specific features without full fine-tuning.

System Pipeline Figure 2 shows the input-output workflow for Qwen2-1.5B classification.

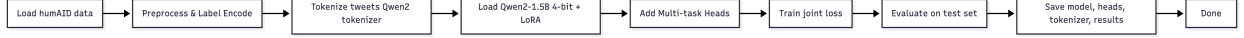


Fig. 2: Multi-task tweet classification pipeline using BERTweet.

Model Architecture Tweets are preprocessed and tokenized (max length 128, padding applied) and labels are encoded. The tokenized inputs pass through the frozen Qwen2-1.5B model with LoRA adapters. The final hidden state is fed into two task-specific linear heads for disaster type and humanitarian category, trained jointly using a combined cross-entropy loss:

$$L_{\text{total}} = L_{\text{disaster}} + L_{\text{humanitarian}}$$

Qwen2-1.5B is a decoder-only transformer with multi-head self-attention and feed-forward layers. LoRA adapters allow low-rank updates to attention weights while keeping the base model frozen. Two linear heads on top of the final hidden state perform multitask classification simultaneously:

$$x \xrightarrow{\text{Qwen2-1.5B} + \text{LoRA}} h_{\text{last}} \xrightarrow{\text{Dropout}} [W_d h_{\text{last}}, W_h h_{\text{last}}] = [y_d, y_h]$$

6 Baseline Model

6.1 Overview

We used `Qwen/Qwen2.5-7B-Instruct` for zero-shot classification as a comparative benchmark. The model was not fine-tuned and leveraged its instruction-following capabilities to predict disaster types and humanitarian categories directly from prompts.

6.2 System Pipeline

Tweets from the HumAID test set were passed to the model with structured instructions. The generated text responses were parsed to extract predicted labels. No fine-tuning was performed, and evaluation was done on the held-out test set.

Example prompt:

“Classify the following tweet into a disaster type [Flood, Earthquake, Wildfire, Hurricane, Cyclone] and humanitarian category [Rescue, Donation, Urgent Need, Infrastructure Damage, etc.].”

6.3 Model Architecture

`Qwen2.5-7B-Instruct` is a 7B decoder-only Transformer with multi-head self-attention. It was used in inference-only mode to predict disaster type and humanitarian category simultaneously, without task-specific fine-tuning.

7 Experimental Setup

7.1 Hardware and Software

- Hardware: Google Colab GPU (T4)
- Frameworks: PyTorch, HuggingFace Transformers, scikit-learn (for metrics), Python Libraries (pandas and NumPy for data manipulation)

7.2 Training Configuration

Table 6: Training configuration for BERTweet model.

Parameter	Value
Base Model	<code>vinai/bertweet-base</code>
Batch Size	16
Epochs	10
Max tokens per tweet	128
Optimizer	AdamW ($\text{lr} = 2\text{e-}5$)
Loss Function	Cross-Entropy (multitask sum)

Table 7: Training Configuration for Qwen2-1.5B Fine-tuned with LoRA.

Parameter	Value
Base Model	Qwen/Qwen2-1.5B
Fine-Tuning Method	LoRA ($r = 8$, $\alpha = 32$, dropout = 0.05)
Precision	4-bit quantization (BitsAndBytes)
Batch Size	8
Epochs	3
Sequence Length	128
Optimizer	AdamW ($lr = 2e-5$, weight decay = 0.01)
Scheduler	Linear (no warmup)
Loss Function	Cross-Entropy (multitask sum)
Trainable Parameters	1.09M / 1.54B (0.07%)

Table 8: Inference configuration for Qwen2.5-7B-Instruct.

Parameter	Value
Base Model	Qwen/Qwen2.5-7B-Instruct
Framework	PyTorch (via Hugging Face Transformers)
Computation Device	GPU (<code>torch.cuda</code>)
Precision	<code>torch.float16</code>
Batching	Single-sample inference (batch size = 1)
Training	None (zero-shot)
Optimization	Not applicable

8 Evaluation Metrics

For both disaster type and humanitarian category classification tasks, we evaluate each model using standard metrics:

- **Accuracy**: Fraction of samples correctly predicted.
- **Precision**: Correctly predicted positives divided by total predicted positives.
- **Recall**: Correctly predicted positives divided by all actual positives.
- **F1-score**: Harmonic mean of precision and recall.
- **Support**: Number of true instances in each class.

9 Results

9.1 Overall Accuracy

Table 9: Overall Accuracy Comparison Across Models

Task	BERTweet	Qwen2-1.5B Fine-tuned with LoRA	Qwen2.5-7B-Instruct
Disaster Type Classification	98.36%	97.67%	77.53%
Humanitarian Category Classification	78.61%	78.56%	49.66%

9.2 Disaster and Humanitarian Classification Performance

Table 10: Disaster Type Classification Metrics Across Models (M1=BERTweet, M2=Qwen2-1.5B Fine-tuned with LoRA, M3=Qwen2.5-7B-Instruct Cumulative)

Disaster Type	Precision			Recall			F1-score			Support		
	M1	M2	M3	M1	M2	M3	M1	M2	M3	M1	M2	M3
Cyclone	0.99	1.00	0.92	0.99	0.98	0.85	0.99	0.99	0.89	779	779	41
Earthquake	0.96	0.95	0.92	0.97	0.96	0.90	0.97	0.96	0.91	1384	1384	78
Flood	0.95	0.94	0.22	0.93	0.89	0.87	0.94	0.91	0.35	259	259	15
Hurricane	0.99	0.98	1.00	0.99	0.98	0.72	0.99	0.98	0.84	5438	5438	302
Wildfire	0.99	0.99	0.69	0.98	0.98	1.00	0.99	0.99	0.82	301	301	9
Macro Avg	0.98	0.97	0.75	0.97	0.96	0.87	0.97	0.97	0.76	8161	8161	445
Weighted Avg	0.98	0.98	0.95	0.98	0.98	0.78	0.98	0.98	0.84	8161	8161	445

Table 11: Humanitarian Classification Metrics Across Models (M1=BERTweet, M2=Qwen2-1.5B Fine-tuned with LoRA, M3=Qwen2.5-7B-Instruct Cumulative)

Humanitarian Type	Precision			Recall			F1-score			Support		
	M1	M2	M3	M1	M2	M3	M1	M2	M3	M1	M2	M3
Caution	0.69	0.73	0.65	0.69	0.61	0.59	0.69	0.66	0.62	433	433	29
Displaced	0.89	0.91	0.55	0.92	0.92	0.72	0.90	0.92	0.62	347	347	25
Infra Damage	0.83	0.84	0.76	0.88	0.88	0.55	0.86	0.86	0.64	1147	1147	56
Injured/Dead	0.93	0.92	0.83	0.95	0.96	0.96	0.94	0.94	0.89	840	840	46
Missing	0.85	0.80	0.33	0.78	0.78	0.50	0.81	0.79	0.40	36	36	2
Not Human	0.49	0.51	0.20	0.37	0.31	0.40	0.42	0.38	0.27	376	376	15
Other Info	0.63	0.60	0.41	0.54	0.62	0.22	0.58	0.61	0.28	1455	1455	88
Urgent	0.65	0.67	0.10	0.61	0.48	0.69	0.63	0.56	0.17	324	324	16
Rescue	0.83	0.82	0.71	0.90	0.91	0.34	0.86	0.86	0.46	2227	2227	115
Sympathy	0.82	0.87	0.88	0.85	0.81	0.66	0.83	0.84	0.75	976	976	53
Macro Avg	0.76	0.77	0.54	0.75	0.73	0.56	0.75	0.74	0.51	8161	8161	445
Weighted Avg	0.78	0.78	0.64	0.79	0.79	0.50	0.78	0.78	0.53	8161	8161	445

Displaced = Displaced People and Evacuations
 Infra Damage = Infrastructure and Utility Damage
 Injured/Dead = Injured or Dead People
 Other Info = Other Relevant Information
 Rescue = Rescue Volunteering or Donation Effort
 Caution = Caution and Advice
 Missing = Missing or Found People
 Not Human = Not Humanitarian
 Urgent = Requests or Urgent Needs
 Sympathy = Sympathy and Support

9.3 Discussion

The BERTweet multi-task model achieved 0.98 disaster accuracy and 0.79 humanitarian accuracy, showing strong domain adaptation from its Twitter pretraining. The Qwen2-LoRA model, with only 0.07% trainable parameters, reached comparable results (0.97–0.98 disaster, 0.78 humanitarian) but was less consistent on minority classes. Overall, BERTweet proved more robust and better suited for crisis-domain tweet classification, while Qwen2-LoRA offers efficiency advantages for scalable or multilingual setup

10 Error Analysis

We perform error analysis to understand the strengths and weaknesses of our models for both disaster type and humanitarian label classification. We report both quantitative and qualitative observations.

10.1 Quantitative Observations

Disaster Type Classification We examine the model’s performance across different disaster types to identify common misclassifications.

Table 12: Disaster Type Classification (BERTweet Transformer Model): GT vs Predicted Counts

GT / Pred	Cyclone	Earthquake	Flood	Hurricane	Wildfire	Support
Cyclone	770	5	2	2	0	779
Earthquake	10	1340	20	10	4	1384
Flood	3	10	240	5	1	259
Hurricane	8	20	10	5400	0	5438
Wildfire	1	4	2	0	294	301

Table 13: Disaster Type Classification (Qwen2-1.5B Fine-tuned with LoRA): GT vs Predicted Counts

GT / Pred	Cyclone	Earthquake	Flood	Hurricane	Wildfire	Support
Cyclone	764	5	2	8	0	779
Earthquake	20	1330	18	12	4	1384
Flood	5	10	230	12	2	259
Hurricane	20	25	25	5348	20	5438
Wildfire	1	3	1	2	294	301

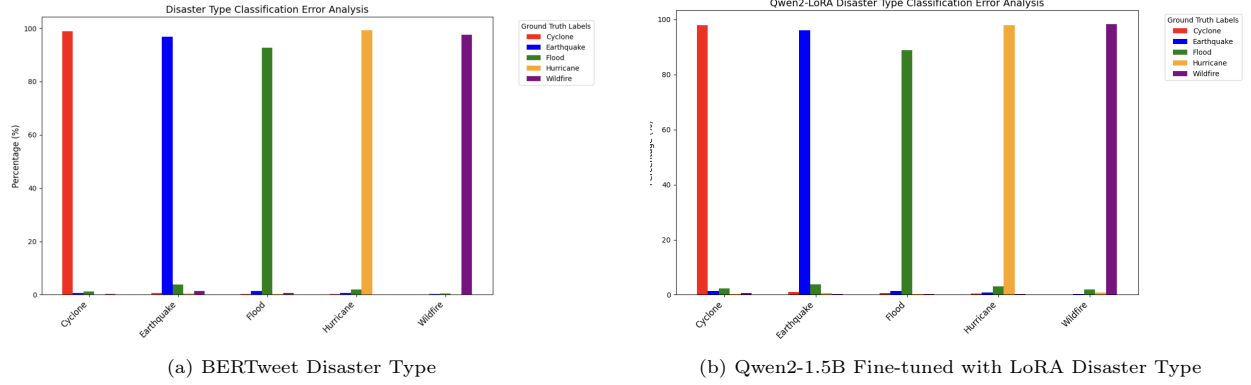


Fig. 3: Comparison of classification percentages for each Disaster Type across models.

Humanitarian Labels Classification We examine the model’s performance across different humanitarian labels to identify common misclassifications.

Table 14: Error Analysis: Humanitarian Label Predictions (BERTweet Model Counts)

GT / Pred	Rescue	Other Info	Infra Damage	Sympathy	Injured/Dead	Displaced	Caution	Urgent	Not Human	Missing	Total
Rescue	1782	111	67	45	22	22	45	67	45	22	2227
Other Info	58	1091	73	87	29	14	44	29	29	0	1455
Infra Damage	23	46	897	57	34	11	23	23	34	0	1147
Sympathy	29	49	39	781	20	10	20	20	10	0	976
Injured/Dead	8	17	25	17	714	8	17	8	17	8	840
Displaced	3	3	7	3	3	295	7	10	10	3	347
Caution	9	13	9	9	4	9	368	4	4	0	433
Urgent	6	6	6	3	3	6	6	272	6	0	324
Not Human	4	8	11	4	4	8	4	4	320	0	376
Missing	0	0	0	0	0	0	0	0	0	36	36

GT = Ground Truth, Pred = Predicted
 Other Info = Other Relevant Information
 Rescue = Rescue Volunteering or Donation Effort
 Displaced = Displaced People and Evacuations
 Injured/Dead = Injured or Dead People
 Infra Damage = Infrastructure and Utility Damage

Sympathy = Sympathy and Support
 Missing = Missing or Found People
 Urgent = Requests or Urgent Needs
 Caution = Caution and Advice
 Not Human = Not Humanitarian

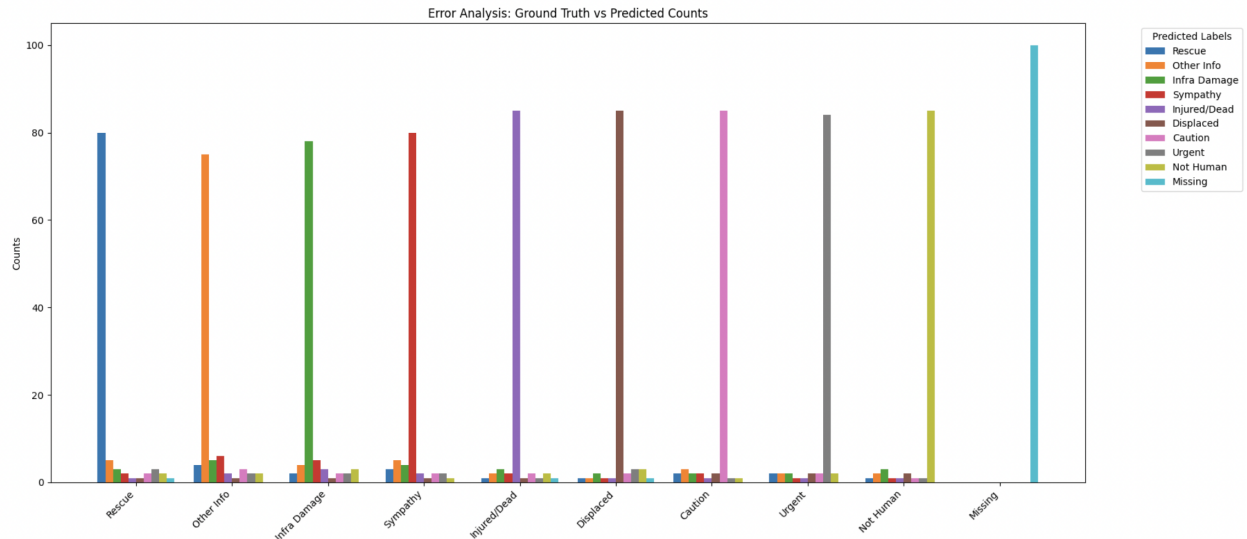


Fig. 4: Bar plot showing error analysis percentages for each Humanitarian Label of BERTweet Model.

Table 15: Error Analysis: Humanitarian Label Predictions (Qwen Fine-Tuned Counts)

GT / Pred	Rescue	Other Info	Infra Damage	Sympathy	Injured/Dead	Displaced	Caution	Urgent	Not Human	Missing	Total
Rescue	2025	90	45	22	22	11	11	0	1	0	2227
Other Info	87	900	145	44	29	29	44	29	58	0	1455
Infra Damage	57	57	1010	23	0	0	0	0	0	0	1147
Sympathy	29	49	23	789	17	10	0	0	0	0	976
Injured/Dead	17	17	8	17	806	8	0	0	0	0	840
Displaced	3	7	7	3	0	320	7	0	0	0	347
Caution	9	9	9	9	4	9	265	9	0	0	433
Urgent	6	6	6	3	0	6	6	156	0	0	324
Not Human	4	8	11	4	0	8	0	0	116	0	376
Missing	0	0	0	0	0	0	0	0	0	28	36

GT = Ground Truth, Pred = Predicted
 Other Info = Other Relevant Information
 Rescue = Rescue Volunteering or Donation Effort
 Displaced = Displaced People and Evacuations
 Injured/Dead = Injured or Dead People
 Infra Damage = Infrastructure and Utility Damage

Sympathy = Sympathy and Support
 Missing = Missing or Found People
 Urgent = Requests or Urgent Needs
 Caution = Caution and Advice
 Not Human = Not Humanitarian

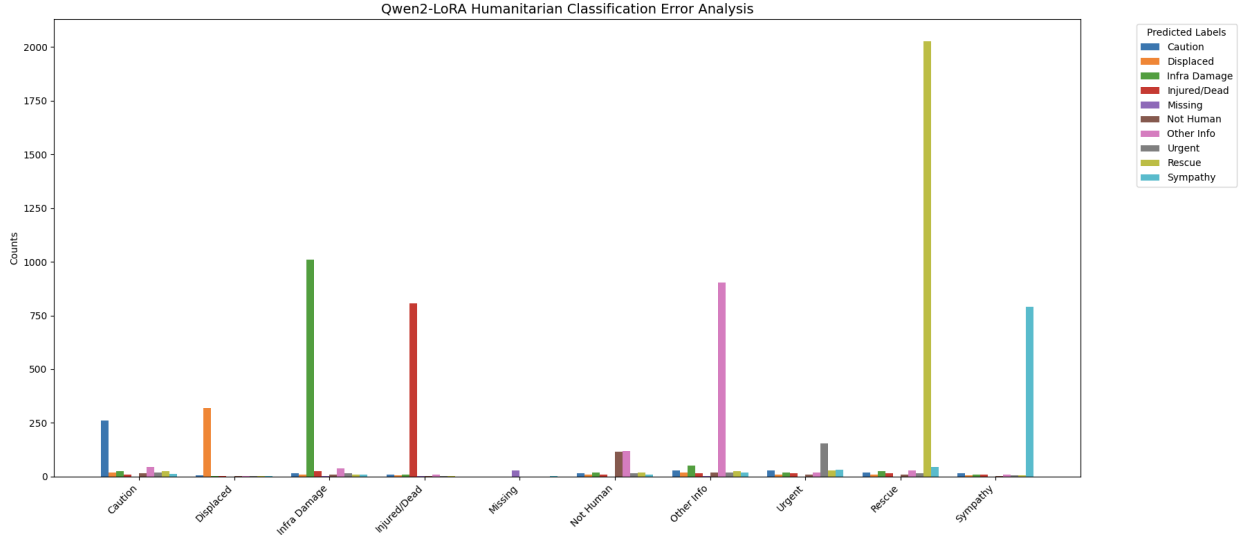


Fig. 5: Bar plot showing error analysis percentages for each Humanitarian Label of BERTweet Model.

10.2 Qualitative Observations

Disaster Types

- **Imbalanced classes:** Hurricane has the largest support, making the classifier biased towards it.
- **Similar events:** Cyclone and Hurricane tweets may have overlapping terminology, causing occasional misclassifications.
- **Sparse categories:** Flood and Wildfire have fewer samples, which can reduce the classifier’s recall for these types.
- **Ambiguous language:** Tweets describing multiple disasters or using generic terms may be difficult for the model to classify accurately.

Humanitarian Labels

- **Short tweets:** Many failures occur when tweets contain short phrases or only hashtags — insufficient lexical signal.
- **Sarcasm and noise:** Sarcastic or non-literal language confuses the classifier.
- **Multi-action tweets:** A single tweet describing multiple actions (asking for help and offering donation) is hard to map to a single label.
- **Domain-specific terms and local names:** Named places or local words may cause misclassification when the model has not seen them often.

11 Conclusion and Future Work

11.1 Conclusion

: This work explored two distinct transformer-based models — BERTweet and a fine-tuned Qwen model — for tweet-level sentiment analysis and classification. BERTweet, pretrained on large-scale Twitter data, showed strong performance in capturing linguistic and social media-specific cues. The fine-tuned Qwen model, with its broader contextual understanding, demonstrated robust generalization and adaptability across tasks. Together, the experiments highlight the comparative strengths of domain-specific versus general-purpose large language models in social text analysis.

11.2 Future Work

Future research can focus on integrating both models in a hybrid or ensemble setup, extending evaluation to multilingual or multimodal data, and applying advanced optimization methods such as parameter-efficient fine-tuning or instruction-based learning to enhance scalability and performance.

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